Policy & Strategy Lab

Instructor Guide, Week 3: Investigate and synthesise

This guide covers Days 11 to 15. Each day follows the same pattern: an overview, the session plan, preparation, a slide-by-slide teaching narrative that matches the day's PowerPoint, likely misconceptions, discussion questions with what strong answers include, full facilitation notes for the activity, the workbook application, assessment, readings and the transition to the next day.

The running case in every demonstration is Madhyapur, a stylised city. Its figures are illustrative and must never be presented as real data. Rules, schemes and institutions named alongside it (the Solid Waste Management Rules, 2016, Swachh Bharat Mission (Urban) 2.0, the Seventh and Twelfth Schedules) are real; check current details before teaching.

Day 11: Finding and judging evidence

Item	Detail
Week	3: Investigate and synthesise (Monday)
Central question	Where is the evidence on our problem, and how far can we trust it?
Session outcome	Each participant has entered and rated at least six sources in your evidence register.
Competencies	E2, E3, J5
Anchor tool	Source reliability assessment
Supporting tools (demonstrated only)	The Indian evidence ecosystem; Citing in Chicago style
Theory link	None today.
Gate	No gate today.

Learning objectives

1. Navigate the main kinds of Indian source: surveys, administrative data, audits, parliamentary material and research.
2. Judge a source by who produced it, how, when, at what level and with what limits.
3. Distinguish what a source establishes from what it only suggests.
4. Cite sources accurately in Chicago style, and never invent a detail.

Session plan

Minutes	Block	Length
0 to 10	Opening problem	10 minutes
10 to 30	Concept: the evidence ecosystem	20 minutes
30 to 45	Demonstration: rating sources	15 minutes
45 to 75	Activity: the evidence hunt	30 minutes
75 to 110	Studio: own evidence register	35 minutes
110 to 120	Debrief and exit task	10 minutes

Preparation and materials

- The Day 11 slide deck (03 Slides)
- Worksheet 5.1 in Workbook 5
- Question cards
- Laptops or phones
- Source rating card

Indian policy context

A scheme dashboard, a CAG performance audit and a newspaper investigation can describe the same programme very differently. Dashboards record what the system reports; audits test records against practice; journalism documents particular places and voices.

Teaching narrative, slide by slide

Slide 1. Title slide. Week 3 is the research week and runs as a studio: short instruction, then supervised working time. Have laptops or phones ready.

Slide 2. Today. Session outcome: by the end of the session each participant has entered and rated at least six sources in your evidence register.

Slide 3. Opening problem. The right answer is: it depends on the question. Collect the lists of what participants would need to know; they will anticipate most of the five questions on the third slide.

Slide 4. The Indian evidence ecosystem. Walk through one row in depth, ideally the one most participants rely on. Before using any figure in teaching material or reports, check the current round and reference year of each survey; several are between rounds.

Slide 5. Five questions to judge any source. These five questions become the reliability column of the evidence register. A rating without a reason drawn from them is not acceptable at Gate 3.

Slide 6. Rating sources for Madhyapur. Two rows are illustrative because Madhyapur is stylised. Participants must locate the real audit and studies relevant to their own state and city. Point out that the low-reliability sources are still useful: the newspaper report documents a mechanism that the official returns cannot.

Slide 7. What a source establishes, and what it only suggests. Read each pair across. The habit is to notice the jump from establishes to suggests. Most overclaiming in policy reports happens at exactly that jump, often in a single word such as shows.

Slide 8. Citing in Chicago style. This rule applies with special force to anything produced with AI help. AI tools invent plausible citations. Every citation in the register must be checked against the original document.

Slide 9. Activity: The evidence hunt. Pods will start with search engines and news. Redirect them to primary sources: the survey report, the audit, the budget document. Praise the pod that finds the original table rather than a summary of it.

Slide 10. Your case: Worksheet 5.1. Individual application, about 20 minutes. Circulate and check that every participant is working on their own case. Spot-check one citation per participant against the original document during the studio block.

Slide 11. What to take away. Ask two participants to state the takeaway they found least obvious, then close with the exit task: Name your most reliable source and its biggest limitation.

Slide 12. Before the next session. Read your two most important sources properly, including the methods section. Write three sentences on what each establishes and what it does not. About 50 minutes.

Likely misconceptions

Misconception	How to respond
Official data are always reliable.	They are often the best available, but administrative data record what the system reports, which can differ from what happened.
Journalism is not evidence.	Credible reporting can document mechanisms and voices that data miss. Judge it by its method and sources, like anything else.
A newer source is a better source.	Check the reference period and the method. An older audit can tell you more than a recent press release.

Discussion questions

Question	What strong answers include
Why might a scheme dashboard and an audit disagree?	Dashboards report what the system records; audits test records against practice. Strong answers explain which question each can answer.
What does representative at state level mean for a district case?	District estimates may be too imprecise to use. Strong answers name an alternative source or state the limitation.

Activity: The evidence hunt

Element	Detail
Purpose	Find and rank credible sources fast, and learn which kinds of source answer which kinds of question.
Scenario	Each pod gets one question and twenty minutes. How has the share of children in a chosen state who can read a Grade 2 text changed over the last decade? Are sanctioned posts in primary health centres in a chosen state filled? What has the CAG said about solid waste management in a chosen state? How much did a chosen state budget for school meals last year, and how much did it spend?
Group size	Pods of four, one question per pod
Time	30 minutes
Materials	Question cards; Laptops or phones; Source rating card
Expected output	Five rated sources and the most useful one, justified

Steps.

1. Find five sources on your question in twenty minutes.
2. For each, record type, producer, period and level.
3. Rate each source's reliability, with a reason.
4. Choose the single most useful source and say why.
5. Report in two minutes; your evidence checker defends one rating.

Facilitation. Pods will start with search engines and news. Redirect them to primary sources: the survey report, the audit, the budget document. Praise the pod that finds the original table rather than a summary of it.

Debrief questions.

- Which source type was most useful for your question, and why?
- Did the most authoritative source answer the question at the right level?
- Where did two sources disagree, and what explains it?

Evaluation criteria	Common errors
Five sources found	Citing a news article about a report instead of the report
Type, period and level recorded	Ignoring the reference year
Reliability justified	Rating by reputation alone

Extension for strong groups. Find the methods note for your best source and identify one limitation its summary did not mention.

Workbook application: Worksheet 5.1, Building the evidence register

Assemble the published evidence your workplan needs, and judge how far each source can be trusted. Spot-check one citation per participant against the original document during the studio block.

Hand in. Updated evidence register by the start of Day 12.

Independent work

Read your two most important sources properly, including the methods section. Write three sentences on what each establishes and what it does not. About 50 minutes.

Assessment

The evidence register is assessed with the final report (15 per cent). Gate 3 feedback uses E2, E3 and J5 and tracker codes EV1, EV2 and EV3.

Readings

- Essential: The methods chapter or note of your main survey or administrative source.
- Optional: The CAG audit reports for your state and sector, searched on the CAG website.

Close and transition

Exit task: Name your most reliable source and its biggest limitation. Transition: Tomorrow: what the numbers actually say. Bring your data.

Day 12: What the numbers say

Item	Detail
Week	3: Investigate and synthesise (Tuesday)
Central question	What does the data actually tell us, and what does it not?
Session outcome	Each participant has produced three charts or tables that disaggregate or compare, each with an interpretation and a limitation.
Competencies	E4, I3, C5, J3
Anchor tool	Disaggregation and comparison in a spreadsheet
Supporting tools (demonstrated only)	Choosing a chart; Correlation and causation; Avoiding false precision
Theory link	None today.
Gate	No gate today.

Learning objectives

1. Compute and read shares, rates, averages and changes correctly.
2. Disaggregate by place, group or time to find where the problem concentrates.
3. Choose a chart that suits the comparison, and label it honestly.
4. Separate what the numbers show from what they suggest.

Session plan

Minutes	Block	Length
0 to 10	Opening problem	10 minutes
10 to 30	Concept: reading numbers	20 minutes
30 to 50	Demonstration: Madhyapur data	20 minutes
50 to 80	Activity: data detective	30 minutes
80 to 110	Studio: own charts	30 minutes
110 to 120	Debrief and exit task	10 minutes

Preparation and materials

- The Day 12 slide deck (03 Slides)
- Worksheet 5.2 in Workbook 5
- The illustrative dataset as a spreadsheet (case bank, case 9)
- Laptops

Indian policy context

District and state averages from surveys and dashboards often hide large differences between blocks, social groups and genders. Disaggregating is usually the single most informative step in an Indian policy analysis.

Teaching narrative, slide by slide

Slide 1. Title slide. Return Gate 2 feedback at the start. Today is a studio: make sure everyone has their main dataset open before the activity.

Slide 2. Today. Session outcome: by the end of the session each participant has produced three charts or tables that disaggregate or compare, each with an interpretation and a limitation.

Slide 3. Opening problem. Collect answers: by ward type, which wards reported, how separation was measured, whether the method changed. The next slides show what the average hid.

Slide 4. Five numbers that are easy to misread. Do the arithmetic on the board for the second point. Ask participants to find one of these five in a source they have already read; most will find at least one.

Slide 5. Data detective: Madhyapur's wards (illustrative data). Work through it: the weighted average uses households as weights. Most of the rise comes from planned colonies. The older core, the largest group, barely moved. The informal settlement figure rests on four of ten wards and may be biased either way. The honest summary: separation improved sharply in planned colonies and hardly at all elsewhere; data for informal settlements are incomplete.

Slide 6. Choosing a chart. Show the Madhyapur table as a sorted bar chart of change by ward group, titled with the finding: separation rose in planned colonies but barely elsewhere. Note that the title carries the argument; the chart supports it.

Slide 7. From a pattern to a claim. This is the Day 4 distinction applied to data. Ask: what comparison would separate the two explanations? Answer: compare wards of similar income with different collection frequency.

Slide 8. False precision. Competency J3. Decimal places signal a precision the data rarely have. A decision-maker trusts an analyst more, not less, when uncertainty is stated plainly.

Slide 9. Activity: Data detective. Do not reveal the traps. Most pods report the average change first. Ask: is this true in every block? Then: did anything change in how the data were collected? The helpline trap is the hardest; it shows that a rise in complaints can mean better access to complain.

Slide 10. Your case: Worksheet 5.2. Individual application, about 20 minutes. Circulate and check that every participant is working on their own case. Ask each participant what their denominator is. Many cannot say at first.

Slide 11. What to take away. Ask two participants to state the takeaway they found least obvious, then close with the exit task: Write the most important number in your case, with its year, its source and one caveat.

Slide 12. Before the next session. Finish your three charts. Ask your pod's evidence checker to review one chart for an honest scale and honest wording. About 50 minutes.

Likely misconceptions

Misconception	How to respond
Numbers are more objective than words.	Numbers embed choices about what to count and how. Read the method.
A big change is a real change.	Check whether the measurement changed, the sample changed or data went missing.
A chart should look impressive.	A chart should make one comparison obvious and honest.

Discussion questions

Question	What strong answers include
Why might a district average improve while the poorest blocks get worse?	Averages weight by size, and gains can concentrate in better-off places. Strong answers propose the disaggregation that would reveal it.

Question	What strong answers include
When is a table better than a chart?	When readers need precise values or there are only a few numbers. Strong answers match the form to the purpose.

Activity: Data detective

Element	Detail
Purpose	Find what a dataset can and cannot support, including the traps hidden in it.
Scenario	Each pod receives an illustrative district dataset for a school meal scheme: enrolment, meals served, days of operation and complaints, by block, for two years. Hidden in it are a misleading average, two blocks with missing months, a change in how meals were counted between the years, and a block where complaints rose after a helpline opened.
Group size	Pods of four
Time	30 minutes
Materials	The illustrative dataset as a spreadsheet (case bank, case 9); Laptops
Expected output	Three observations, two interpretations, one limitation

Steps.

1. Compute the district figure for each year, then the block figures.
2. Find where the average hides a different story.
3. Identify every missing or inconsistent value.
4. Write three observations, two interpretations and one limitation.
5. Present the claim you are most, and least, confident about.

Facilitation. Do not reveal the traps. Most pods report the average change first. Ask: is this true in every block? Then: did anything change in how the data were collected? The helpline trap is the hardest; it shows that a rise in complaints can mean better access to complain.

Debrief questions.

- What did the average hide?
- Which trap did your pod miss, and why?
- What claim would you put in front of a District Collector, and how would you word it?

Evaluation criteria	Common errors
Disaggregation done	Reporting only the average
Traps identified	Missing the change in counting
Claims worded at the right strength	Treating more complaints as worse service

Extension for strong groups. Build one honest chart of the hidden story, with year, source and a note on missing data.

Workbook application: Worksheet 5.2, Data analysis

Find where your problem concentrates and what the numbers can, and cannot, support. Ask each participant what their denominator is. Many cannot say at first.

Hand in. Worksheet 5.2 by the start of Day 13.

Independent work

Finish your three charts. Ask your pod's evidence checker to review one chart for an honest scale and honest wording. About 50 minutes.

Assessment

Feeds Gate 3. Feedback uses E4, I3, C5 and J3 and tracker codes CA2 and EV2.

Readings

- Essential: Spiegelhalter, David. The Art of Statistics: Learning from Data. London: Pelican, 2019. Chapter 1.
- Optional: Schwabish, Jonathan. Better Data Visualizations. New York: Columbia University Press, 2021. The chapters on choosing a chart.

Close and transition

Exit task: Write the most important number in your case, with its year, its source and one caveat. Transition: Tomorrow: what people's experience, and other places' attempts, can tell us.

Day 13: Experience and comparison

Item	Detail
Week	3: Investigate and synthesise (Wednesday)
Central question	What can people's experience and other places' attempts tell us that numbers cannot?
Session outcome	Each participant has built a comparative case matrix and drawn themes from published qualitative sources.
Competencies	E5, E6
Anchor tool	Comparative case matrix
Supporting tools (demonstrated only)	Coding published qualitative evidence into themes; Conditions for policy transfer
Theory link	None today.
Gate	No gate today.

Learning objectives

1. Draw themes from published qualitative sources such as audits, social audits, testimony and field studies.
2. Compare how two or more places tackled a similar problem.
3. Identify the conditions that made an approach work, and which of them transfer.
4. Use qualitative evidence to test mechanisms, not just to illustrate.

Session plan

Minutes	Block	Length
0 to 10	Opening problem	10 minutes
10 to 30	Concept: qualitative evidence	20 minutes
30 to 50	Demonstration: comparing cities	20 minutes
50 to 80	Activity: policy transfer challenge	30 minutes
80 to 110	Studio: own comparison	30 minutes
110 to 120	Debrief and exit task	10 minutes

Preparation and materials

- The Day 13 slide deck (03 Slides)
- Worksheet 5.3 in Workbook 5
- Two-state comparison cards (stylised)
- Comparative case matrix template

Indian policy context

Indian cities have tried different routes to separated waste collection: Indore's citywide enforcement and campaigns, Pune's partnership with a waste pickers' cooperative, and Ambikapur's centres run by women's self-help groups. Each worked under particular conditions.

Teaching narrative, slide by slide

Slide 1. Title slide. Remind participants that this course uses published qualitative evidence only: no interviews, surveys or observation.

Slide 2. Today. Session outcome: by the end of the session each participant has built a comparative case matrix and drawn themes from published qualitative sources.

Slide 3. Opening problem. Indore's repeated top placing in the Swachh Survekshan rankings is widely reported; verify the years before citing. Collect the lists: leadership, budget, enforcement, size, the role of waste pickers, processing capacity. These become the rows of the comparative matrix.

Slide 4. What published qualitative evidence can do. Without fieldwork, published qualitative evidence is how participants reach mechanisms and voices. Social audit reports of the rural employment guarantee, CAG performance audits and standing committee testimony are especially rich Indian sources.

Slide 5. Coding sources into themes (illustrative, for Madhyapur). Coding means reading each source with the hypothesis register beside you and recording where each theme appears. Stress independence: three newspapers repeating one press release count as one source.

Slide 6. A comparative case matrix. Pune's partnership with the SWaCH cooperative and Ambikapur's model run by women's self-help groups are well documented in published case studies. Use the matrix to ask, row by row, whether the conditions exist in Madhyapur. The last row is the transfer judgement, not a recommendation yet.

Slide 7. Policy transfer: what travels and what does not. The policy transfer literature, including Dolowitz and Marsh, distinguishes copying, emulation, combination and inspiration. In practice, principles and processes travel better than whole programmes.

Slide 8. Copying the form without the function. Ask participants to name the function behind the most visible form in their own comparison case. If they cannot, the transfer judgement is not yet ready.

Slide 9. Activity: The policy transfer challenge. Pods tend to treat the intervention as one package. Push them to separate the components, because some travel and others do not. Biometric attendance is the classic example of a form that can be copied without the supervision that makes it work.

Slide 10. Your case: Worksheet 5.3. Individual application, about 20 minutes. Circulate and check that every participant is working on their own case. Help participants find at least one social audit, CAG report or committee report for their sector; these are often the best published qualitative sources.

Slide 11. What to take away. Ask two participants to state the takeaway they found least obvious, then close with the exit task: Name the condition that most limits transfer to your case.

Slide 12. Before the next session. Prepare for the red team: write your three strongest findings so far, each with its two best sources. About 50 minutes.

Likely misconceptions

Misconception	How to respond
Qualitative evidence is anecdote.	Systematically coded evidence from several independent sources is evidence about mechanisms.
Best practice can be copied.	Practice that worked somewhere depended on conditions. Transfer the principle; adapt the form.
A famous success is proven.	Check what the evidence actually shows, including any evaluations, before treating it as a model.

Discussion questions

Question	What strong answers include
How many sources make a theme credible?	There is no fixed number, but independence matters more than count. Strong answers distinguish three newspapers repeating one report from three independent studies.
What is the difference between copying a form and copying a function?	A form is the visible structure; a function is what it achieves through people's behaviour. Strong answers give an example from their case.

Activity: The policy transfer challenge

Element	Detail
Purpose	Judge whether an intervention that succeeded elsewhere would work in a different context.
Scenario	A state wants to copy a neighbouring state's apparent success in reducing teacher absence through biometric attendance, community monitoring and fewer non-teaching duties for teachers. Each pod compares the two stylised states on the conditions that made it work in the first.
Group size	Pods of four
Time	30 minutes
Materials	Two-state comparison cards (stylised); Comparative case matrix template
Expected output	A transfer judgement for each component

Steps.

1. List the components of the intervention.
2. For each, list the conditions it depended on in the original state.
3. Check whether those conditions exist in the adopting state.
4. Classify each component: transfers, transfers with adaptation, or does not transfer.
5. Write one paragraph on what to copy and what to build first.

Facilitation. Pods tend to treat the intervention as one package. Push them to separate the components, because some travel and others do not. Biometric attendance is the classic example of a form that can be copied without the supervision that makes it work.

Debrief questions.

- Which component depended most on local conditions?
- What would you need to build before copying?
- How would you know within six months whether the transfer was working?

Evaluation criteria	Common errors
Components separated	Copying the whole package
Conditions identified for each	Ignoring capability and leadership
A justified transfer judgement	Assuming success elsewhere is proven

Extension for strong groups. Find a published evaluation of a real intervention of this kind and test your judgement against it.

Workbook application: Worksheet 5.3, Comparative and qualitative evidence

Learn from other places and from published accounts of people's experience, and judge what transfers to your case. Help participants find at least one social audit, CAG report or committee report for their sector; these are often the best published qualitative sources.

Hand in. Worksheet 5.3 by the start of Day 14.

Independent work

Prepare for the red team: write your three strongest findings so far, each with its two best sources. About 50 minutes.

Assessment

Feeds Gate 3. Feedback uses E5 and E6 and tracker code EV1.

Readings

- Essential: Dolowitz, David P., and David Marsh. "Learning from Abroad: The Role of Policy Transfer in Contemporary Policy-Making." Governance 13, no. 1 (2000).
- Optional: Andrews, Pritchett and Woolcock, Building State Capability (2017), on isomorphic mimicry, revisited.

Close and transition

Exit task: Name the condition that most limits transfer to your case. Transition: Tomorrow your hypotheses face the evidence, and a red team.

Day 14: Testing the hypotheses

Item	Detail
Week	3: Investigate and synthesise (Thursday)
Central question	Which of our explanations survive the evidence?
Session outcome	Each participant has recorded a verdict for each hypothesis with its evidence, and responded in writing to a red team.
Competencies	F3, F5, E6, J4
Anchor tool	Hypothesis testing matrix
Supporting tools (demonstrated only)	Claim, evidence and warrant; Red team review
Theory link	None today.
Gate	No gate today.

Learning objectives

1. Weigh supporting and disconfirming evidence for each hypothesis.
2. Make arguments checkable with claim, evidence and warrant.
3. Record verdicts honestly: supported, weakened or unresolved.
4. Respond to challenge without overstating.

Session plan

Minutes	Block	Length
0 to 10	Opening problem	10 minutes
10 to 25	Concept: weighing evidence	15 minutes
25 to 40	Demonstration: Madhyapur verdicts	15 minutes
40 to 85	Thursday challenge: red team review	45 minutes
85 to 110	Own case: testing matrix	25 minutes
110 to 120	Debrief and exit task	10 minutes

Preparation and materials

- The Day 14 slide deck (03 Slides)
- Worksheet 6.1 in Workbook 6
- Red team question cards
- Timer
- Pod log

Indian policy context

Evaluations of Indian programmes often find that the favoured explanation was partly right and a rival partly right too. Reporting that honestly is more useful to a decision-maker than a single tidy cause.

Teaching narrative, slide by slide

Slide 1. Title slide. Third Thursday challenge. Set the tone carefully before the red team: criticism is quality control, not a personal attack.

Slide 2. Today. Session outcome: by the end of the session each participant has recorded a verdict for each hypothesis with its evidence, and responded in writing to a red team.

Slide 3. Opening problem. The answer is the strongest disconfirming evidence and the best evidence for the rival. This is the Day 9 discipline applied now that evidence is in hand.

Slide 4. The hypothesis testing matrix: Madhyapur (illustrative). H2 being weakened is an important moment in the spine case: the bins-for-all fix that many cities favour would not have worked on its own. Compare with the sealed Day 3 answer: H1 survived, but the vehicle rival now has to be part of the diagnosis.

Slide 5. Claim, evidence, warrant. This structure comes from Stephen Toulmin's model of argument. Ask a participant to state their strongest finding in all three parts; usually the warrant is missing or is a restatement of the claim.

Slide 6. Three verdicts, honestly recorded. Competency F5 rewards updating on evidence. Participants who weaken or drop a favoured hypothesis should be told explicitly that this counts in their favour.

Slide 7. Red team rules. These five questions are the course's standard red team set and will return on Day 20 in the defence. Make the red team ask questions, not give speeches.

Slide 8. Responding to challenge. Model a concession yourself before the activity: take a challenge to the Madhyapur H1 verdict and answer, that is fair, the evidence comes from cities with different vehicles, so I will say supported in comparable cities rather than supported.

Slide 9. Activity: Red team review. The red team's job is to improve the analysis. Stop any challenge that becomes personal. Reward concessions publicly: a presenter who says that is right, I will weaken the claim is modelling the skill.

Slide 10. Your case: Worksheet 6.1. Individual application, about 20 minutes. Circulate and check that every participant is working on their own case. Look for registers where every hypothesis is supported; ask what evidence was searched for that could have weakened them.

Slide 11. What to take away. Ask two participants to state the takeaway they found least obvious, then close with the exit task: Write down the hypothesis whose verdict surprised you most.

Slide 12. Before the next session. Revise your diagnosis in the light of the verdicts. Draft the governing message for tomorrow's memo in one sentence. About 50 minutes.

Likely misconceptions

Misconception	How to respond
A supported hypothesis is proven.	It has survived the tests you could run. State what further evidence would strengthen or overturn it.
Conceding a point weakens the analysis.	It strengthens credibility. Overstating weakens it.
Unresolved means the work failed.	It means the evidence does not settle the question, which the decision-maker needs to know.

Discussion questions

Question	What strong answers include
Why is the warrant the most common gap?	Analysts assume the link between evidence and claim is obvious. Strong answers show a case where the same evidence supports two different claims.

Question	What strong answers include
How should confidence be expressed to a decision-maker?	In plain words tied to the evidence, not false precision. Strong answers draft the sentence they would use.

Activity: Red team review

Element	Detail
Purpose	Stress-test each analyst's strongest finding, and practise responding to challenge without overstating.
Scenario	Each participant presents their strongest finding in three minutes: claim, evidence and warrant. A red team from another pod challenges it with the five red team questions. The presenter then has two minutes to respond and must state any change to the claim.
Group size	Two pods pair up and red-team each other
Time	45 minutes (Thursday challenge)
Materials	Red team question cards; Timer; Pod log
Expected output	A revised claim for each participant, in the pod log

Steps.

1. Presenter: claim, evidence and warrant in three minutes.
2. Red team: the five questions, one minute each, no speeches.
3. Presenter: respond in two minutes, conceding where right.
4. Scribe: record each challenge and response in the pod log.
5. Presenter: write the revised claim before leaving.

Facilitation. The red team's job is to improve the analysis. Stop any challenge that becomes personal. Reward concessions publicly: a presenter who says that is right, I will weaken the claim is modelling the skill.

Debrief questions.

- Which challenge changed a claim most?
- Which question was hardest to answer, and why?
- What did you learn about your own evidence from watching another analyst's?

Evaluation criteria	Common errors
Claim, evidence and warrant all stated	Defending every point
Challenges are specific	Challenges that are opinions without reasons
Responses concede where appropriate	Revised claims that become vaguer rather than more precise

Extension for strong groups. Red-team your own decision-maker's most likely objection: write it down and answer it.

Workbook application: Worksheet 6.1, Hypothesis testing matrix

Decide which explanations survive the evidence, and say how confident you are. Look for registers where every hypothesis is supported; ask what evidence was searched for that could have weakened them.

Hand in. Worksheet 6.1 by the start of Day 15, for Gate 3.

Independent work

Revise your diagnosis in the light of the verdicts. Draft the governing message for tomorrow's memo in one sentence. About 50 minutes.

Assessment

Gate 3 (Day 15). Feedback uses F3, F5, E6 and J4 and tracker codes HY1 and JU1.

Readings

- Essential: Toulmin, Stephen. The Uses of Argument. Cambridge: Cambridge University Press, 1958. The chapter on the layout of arguments.
- Optional: Heuer, Psychology of Intelligence Analysis (1999), chapter 8, revisited.

Close and transition

Exit task: Write down the hypothesis whose verdict surprised you most. Transition: Tomorrow you turn the evidence into a two-page memo with a governing message, and submit Gate 3.

Day 15: Synthesis and the diagnostic memo

Item	Detail
Week	3: Investigate and synthesise (Friday)
Central question	How do we turn a body of evidence into an argument a decision-maker can use?
Session outcome	Each participant has written a two-page diagnostic memo with a governing message, and submitted Gate 3.
Competencies	I1, F4, J3
Anchor tool	Pyramid Principle with SCQA
Supporting tools (demonstrated only)	The executive summary; Rumelt's kernel of strategy
Theory link	Richard Rumelt: Does your argument contain a diagnosis, a guiding policy and coherent action, or is it a list of goals?
Gate	Gate 3 is due today.

Learning objectives

1. Distinguish findings from insights.
2. Structure an argument top-down with the Pyramid Principle.
3. Open with Situation, Complication, Question and Answer.
4. Keep facts, inferences and assumptions visibly separate in the memo.

Session plan

Minutes	Block	Length
0 to 10	Opening problem	10 minutes
10 to 25	Concept: answer first	15 minutes
25 to 40	Demonstration: Madhyapur pyramid	15 minutes
40 to 60	Activity: governing message race	20 minutes
60 to 100	Gate 3: pod peer review	40 minutes
100 to 120	Debrief and planning	20 minutes

Preparation and materials

- The Day 15 slide deck (03 Slides)
- Worksheet 6.2 in Workbook 6
- Evidence dossier (case bank, case 10)
- Pyramid template

Indian policy context

Senior Indian officials typically read briefs under severe time pressure, often between meetings. A note whose first paragraph states the answer and the decision required is far more likely to be acted on than one that builds slowly to its point.

Teaching narrative, slide by slide

Slide 1. Title slide. Gate day. Gate 3 covers Worksheets 5.1 to 5.3, 6.1 and today's memo. The memo is the centrepiece.

Slide 2. Today. Session outcome: by the end of the session each participant has written a two-page diagnostic memo with a governing message, and submitted Gate 3.

Slide 3. Opening problem. Most participants' recent writing starts with background. The contrast makes the point of the session: decision-makers read the start, so the answer goes there.

Slide 4. Findings are not insights. An insight is a claim that connects findings and has implications for action. It is still tagged: in the Madhyapur case it is an inference from several findings, with medium confidence.

Slide 5. The Pyramid Principle: answer first, then support. Barbara Minto developed the Pyramid Principle at McKinsey. The three key lines must answer the question the governing message raises (why?) and should not overlap: the MECE test from Day 3 applies here too.

Slide 6. SCQA: the opening of the Madhyapur memo. The situation should be uncontroversial, something the reader already accepts. The complication creates the tension. The question is the one the reader would ask. The answer is the governing message.

Slide 7. Writing the memo: five rules. The memo is diagnostic: it establishes what the problem is and why. It ends by setting up the decision for Week 4, not by recommending an option yet.

Slide 8. A thinker for your case. Richard Rumelt, Good Strategy/Bad Strategy, 2011. Rumelt argued that the kernel of good strategy is a diagnosis that names the critical challenge, a guiding policy for dealing with it, and coherent actions that carry the policy out. Put the question to the room: Does your argument contain a diagnosis, a guiding policy and coherent action, or is it a list of goals? Then ask each participant to answer it for their own case in one sentence.

Slide 9. Activity: The governing message race. Pods will reach different governing messages from the same evidence. That is the point of the comparison: ask which message is best supported, not which is most striking. Watch for messages that are topic headings rather than claims.

Slide 10. Gate 3 pod peer review: 40 minutes. The first-paragraph test is the quickest check of a memo. If the reviewer cannot state the answer from the first paragraph, the memo is not yet top-down.

Slide 11. Your case: Worksheet 6.2. Individual application, about 20 minutes. Circulate and check that every participant is working on their own case. Apply the first-paragraph test to every memo you see during the session.

Slide 12. What to take away. Ask two participants to state the takeaway they found least obvious, then close with the exit task: Read your governing message to a neighbour. Write down their first question.

Slide 13. Before the next session. Rest. If you have time, read your memo aloud and cut every sentence the decision-maker would skip. About 30 minutes.

Likely misconceptions

Misconception	How to respond
A memo should build up to its conclusion.	Decision-makers read the start. Put the answer first.
More findings make a stronger memo.	Selection is the skill. Include what supports or qualifies the governing message.

Misconception	How to respond
Uncertainty weakens a memo.	Stated clearly and briefly, it strengthens trust.

Discussion questions

Question	What strong answers include
Why do analysts bury the conclusion?	Habits from academic writing, and fear of being wrong. Strong answers show how the Ledger lets them state a conclusion with its confidence.
How is a diagnostic memo different from a recommendation?	It establishes what the problem is and why; the recommendation chooses between options. Strong answers say what decision the memo sets up.

Activity: The governing message race

Element	Detail
Purpose	Reduce a body of evidence to one defensible governing message and three key lines.
Scenario	Each pod gets the same short evidence dossier on a common case: low take-up of a maternity benefit scheme in a stylised district, with six findings of mixed strength. In fifteen minutes, write the governing message and three key lines, then compare across pods.
Group size	Pods of four
Time	20 minutes
Materials	Evidence dossier (case bank, case 10); Pyramid template
Expected output	A governing message and three key lines

Steps.

1. Read the six findings and tag each as fact, inference or assumption.
2. Group them into three key lines that do not overlap.
3. Write the governing message the three key lines support.
4. Check: does any key line rest on an assumption alone?
5. Post your pyramid and compare it with the other pods'.

Facilitation. Pods will reach different governing messages from the same evidence. That is the point of the comparison: ask which message is best supported, not which is most striking. Watch for messages that are topic headings rather than claims.

Debrief questions.

- Why did pods reach different messages from the same evidence?
- Which governing message would best survive a sceptical official?
- Which finding did every pod leave out, and was that right?

Evaluation criteria	Common errors
The governing message is a claim, not a topic	Topic headings instead of claims
Key lines do not overlap	Overlapping key lines
Each key line rests on evidence	A message stronger than its evidence

Extension for strong groups. Rewrite the governing message for a different decision-maker: the state department rather than the district.

Workbook application: Worksheet 6.2, Diagnostic memo

Turn your evidence into a two-page argument a decision-maker can act on. Apply the first-paragraph test to every memo you see during the session.

Hand in. Gate 3 (Worksheets 5.1 to 5.3 and 6.1, revised, and the memo) by the end of Day 15. Feedback returned by Day 17.

Independent work

Rest. If you have time, read your memo aloud and cut every sentence the decision-maker would skip. About 30 minutes.

Assessment

Gate 3: detailed feedback on domains E and F and on I1, under the five headings. Formative.

Readings

- Essential: Minto, The Pyramid Principle, the chapters on the pyramid structure and on introductions (edition to confirm).
- Optional: Rumelt, Good Strategy/Bad Strategy (2011), the chapter on the kernel of good strategy.

Close and transition

Exit task: Read your governing message to a neighbour. Write down their first question. Transition: Week 4 begins: generating options without jumping to the first attractive idea.